What 6,451 exoplanets tell us

Total number of planets
6,451

Number of evaluable planets
836

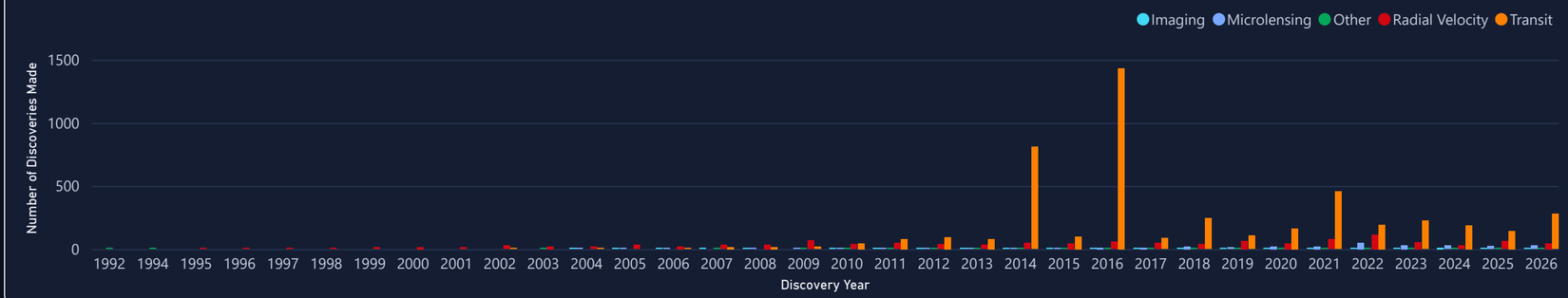
Habitable Zone Candidates
10

Key takeaways

- Transit surveys account for about 74% of the 6,451 known planets, and the 2014 and 2016 spikes come from Kepler's batch validations (715 and 1,284 planets).
- About 35% of planets with a measured radius are 2.0 Earth radii or smaller, and Neptune-sized planets are the largest group at 43%.
- Only 10 of 836 planets with measured insolation and radius meet your habitable-zone and size criteria.
- 8 of those 10 orbit M dwarfs, which host only 6.9% of all planets. That's most likely detection bias, not a real pattern.

Exoplanet Discoveries by Year

Radial Velocity: 312 discoveries before 2010 vs. 60 for Transit. Transit: 4,679 after 2010 vs. 855 for Radial Velocity

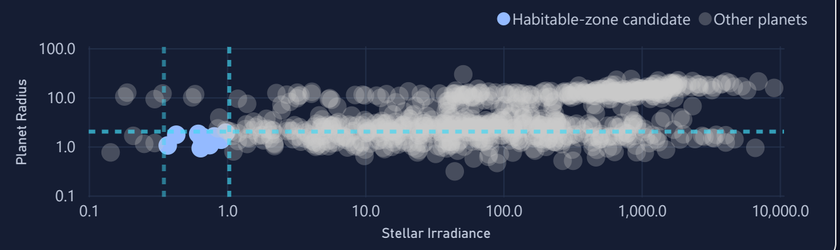


10 Candidates

Planet Name	Insolation (Earth = 1)	Radius (Earth radii)
TRAPPIST-1 f	0.37	1.05
TRAPPIST-1 e	0.65	0.92
TOI-715 b	0.67	1.55
TOI-2094 b	0.98	1.90
LP 890-9 c	0.91	1.37
LHS 1140 b	0.43	1.73
Kepler-442 b	0.66	1.34
Kepler-296 f	0.62	1.80
Kepler-1652 b	0.81	1.60
Kepler-1649 c	0.75	1.06

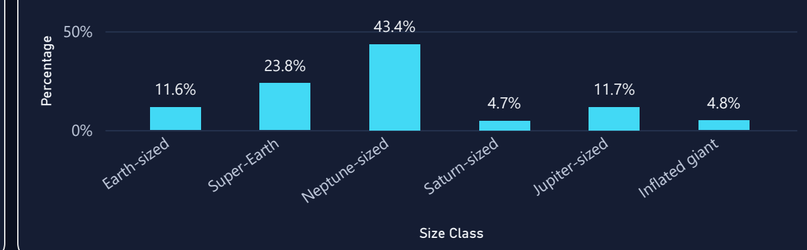
Planet Size vs. Stellar Irradiance

Insolation between 0.35 and 1.04 times Earth's, radius up to 2.0 Earth radii



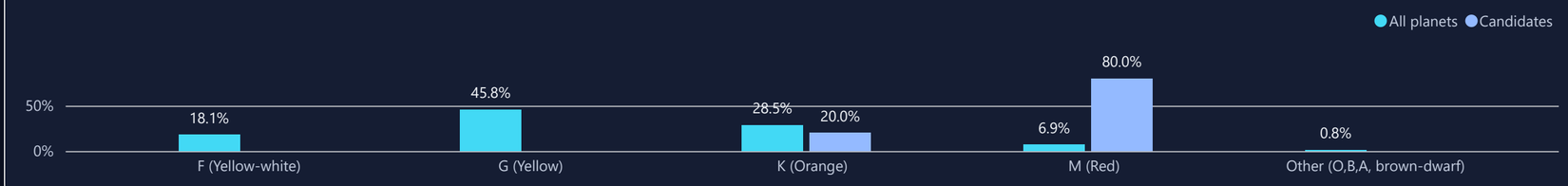
Exoplanets by Size Class

Share of the 4,826 planets with a measured radius



Distribution of Star Types

Share of planets by host star class, all 5,678 planets with a known temperature vs. the 10 candidates



Discovery Year

1992 — 2026

Planet Size Class

- [ ] Earth-sized
- [ ] Super-Earth
- [ ] Neptune-sized
- [ ] Saturn-sized
- [ ] Jupiter-sized
- [ ] Inflated giant